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**Master of Computer Applications
Semester – III
Data Analytics for Business
Experiment No.: 09**

Title: Exploratory Data Analysis on Business Datasets.

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Experiment No.: 09

Exploratory Data Analysis on Business Datasets.

Problem Statement

You are a Data Analyst at a consulting firm. A client has provided a large business dataset and wants to understand key trends and patterns. Perform exploratory data analysis (EDA) to uncover meaningful business insights and identify opportunities for improvement.

Methodology / Steps Followed

Step 1: Data Collection

- Obtained sales data from Kaggle datasets.

Step 2: Data Import

- Opened Power BI Desktop.
- Imported all datasets using the **Get Data** option.
- Loaded CSV file into Power BI.

Step 3: Data Cleaning and Transformation

- Removed duplicate records.
- Handled missing values.
- Converted data types where necessary

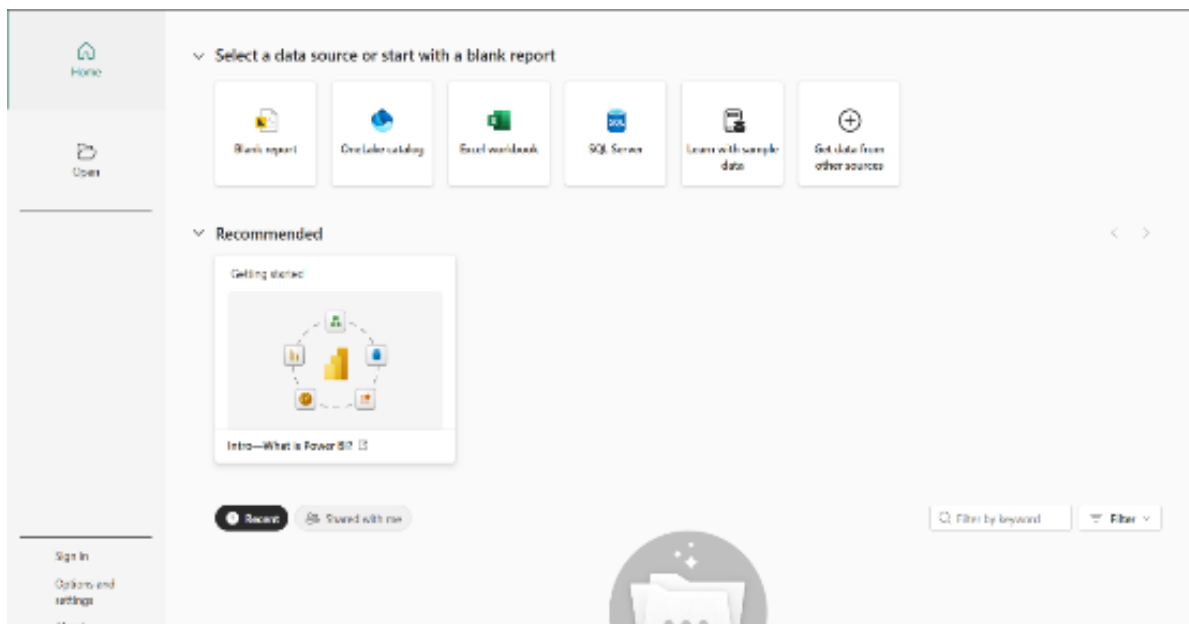
Step 4: Dashboard Development

- Created KPI Cards to display Total Revenue, Total Orders, Unique Customers, Total Units Sold, and Average Customer Rating for an overall view of e-commerce performance.
- Developed a Bar Chart to compare Revenue across different Regions, helping to identify the regions contributing the highest sales.
- Created a Column Chart to analyse Revenue across different Product Categories, highlighting the performance of Electronics, Clothing, Home, and Beauty categories.
- Developed a Line Chart to visualize the year-wise revenue trend, allowing changes in sales performance over different years to be analysed.
- Created a Donut Chart to analyse the distribution of Revenue across different Payment Methods, including Card, COD, and Wallet.
- Developed a Bar Chart to compare Revenue across Product Categories, providing a clear view of category-level sales performance.
- Created a Column and Line Combination Chart to compare Revenue and Order Distribution across different Regions and identify regional differences in business performance.

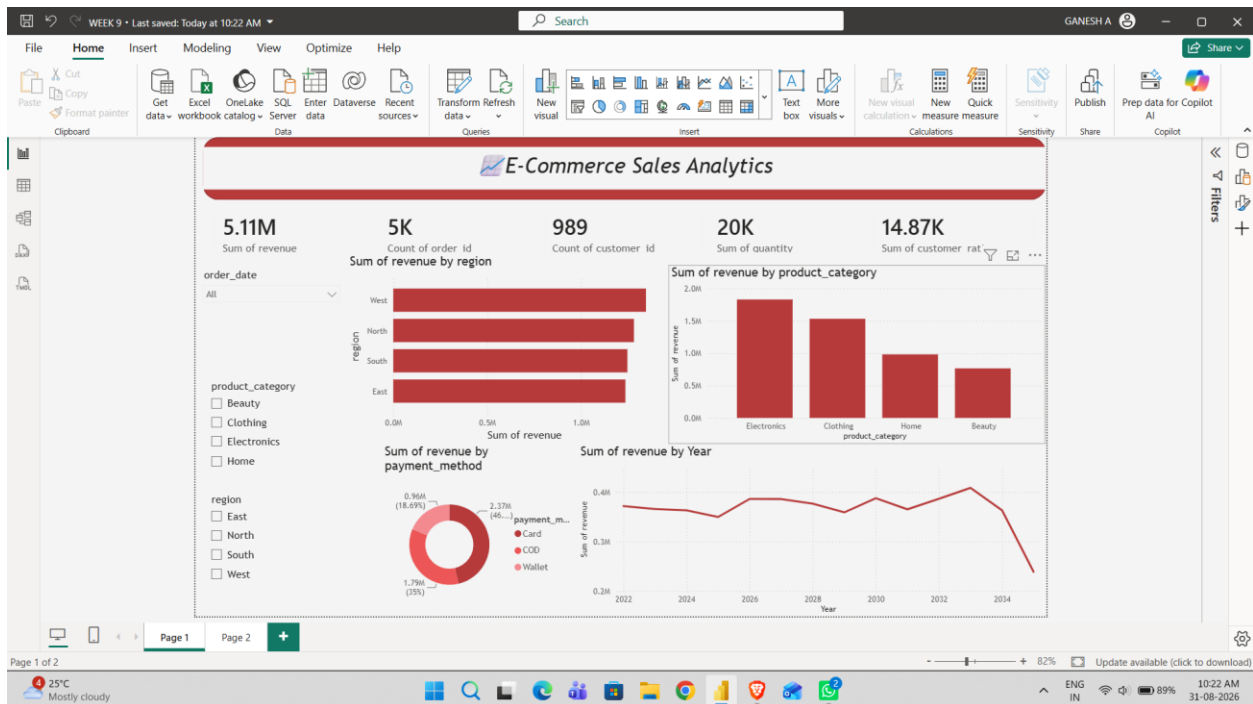
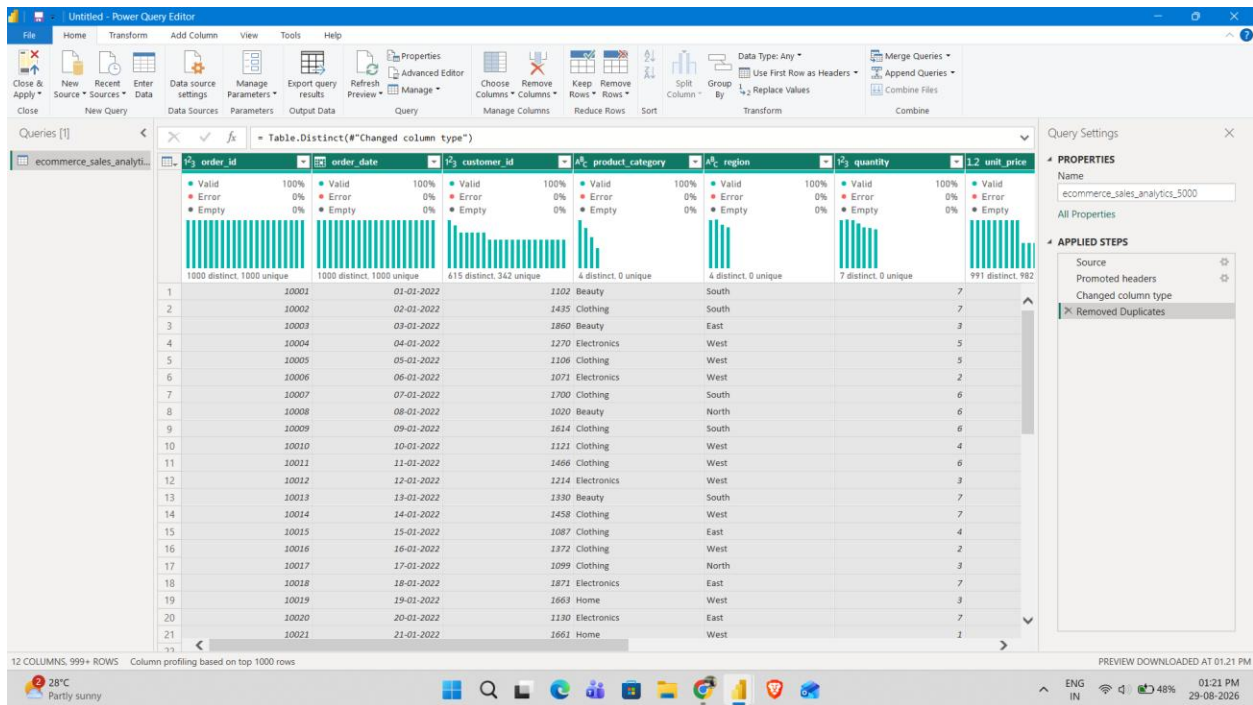
- Developed a Bar Chart to compare Revenue across different Payment Methods, highlighting the contribution of Card, COD, and Wallet transactions.
- Added Interactive Slicers for Order Date, Product Category, and Region to enable dynamic filtering and interactive exploration of the dashboard.
- Designed the dashboard with a two-page layout, separating overall sales performance from detailed customer and business analysis for better readability and interpretation

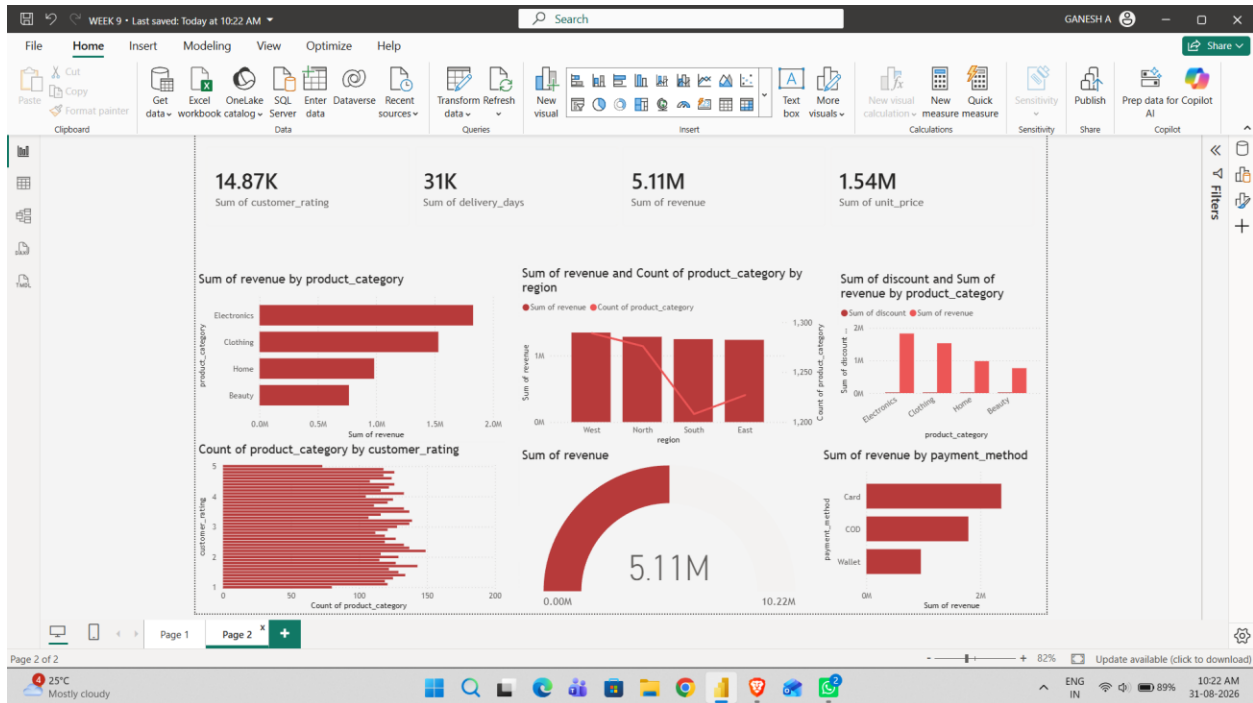
Output

- Opened Power BI file.



- Loaded the dataset using Get Data /CSV.
- Transform The data.
- Applied cleaning using power Query And Applied cleaning steps where ever necessary.





Result.

- The dashboard provides a clear overview of the e-commerce business performance, covering 5,000 orders across different products, regions, and payment methods.
- The business generated approximately 5.11 million in total revenue, indicating a strong level of sales activity during the analysed period.
- Around 20K units were sold, showing consistent customer demand across the different product categories.
- The analysis identified 989 unique customers, providing a useful view of the customer base and order activity.
- Electronics generated the highest revenue, followed by Clothing, Home, and Beauty, making Electronics the strongest-performing product category.
- West recorded the highest regional revenue, while the other regions showed relatively comparable performance, indicating a fairly balanced regional market.
- Card payments contributed the highest revenue, followed by COD and Wallet, showing that card transactions were the most significant payment method.
- The revenue trend remained relatively stable across the analysed years, with some fluctuations in yearly performance.

- Customer ratings provide an indication of overall customer satisfaction, which can be further analysed alongside delivery time and product category.
- The dashboard's interactive slicers allow users to analyse sales performance by date, product category, and region, making it easier to identify specific business trends and patterns.

Conclusion

The E-Commerce Sales Analytics Dashboard provides a clear and interactive view of overall sales performance and customer-related patterns. By analysing revenue, orders, customers, product categories, regions, payment methods, pricing, and customer ratings, the dashboard helps identify important business trends and areas of performance. The analysis highlights differences in category and regional sales while providing a better understanding of customer purchasing behaviour. The interactive slicers make it easy to explore specific categories, regions, and time periods. Overall, the dashboard converts raw e-commerce data into meaningful business insights that can support better sales planning, performance evaluation, and data-driven decision-making.

Learning Outcomes

Through this project, I learned:

- ☐ Gained practical experience in Exploratory Data Analysis (EDA) using a real-world e-commerce sales dataset.
- ☐ Learned to clean, transform, and prepare data using Power Query in Power BI.
- ☐ Developed skills in creating interactive dashboards and KPI cards to monitor business performance.
- ☐ Learned to analyse sales trends across product categories, regions, and payment methods.
- ☐ Gained experience in selecting and designing appropriate charts and visualizations for different business questions.
- ☐ Learned to analyse customer behaviour, purchasing patterns, and sales performance using data-driven visuals.
- ☐ Improved understanding of how to identify business trends and meaningful insights from raw datasets.
- ☐ Developed skills in using interactive slicers and filters to perform dynamic analysis.
- ☐ Gained practical knowledge of presenting analytical findings in a clear, professional, and user-friendly dashboard.

☐ Improved ability to use data insights to support business decision-making and performance evaluation.

Github Link:

https://github.com/GANESH-A08/Data_Analytics_projects/tree/main/Week%209%20Experiment